

IoT Challenge #3 - Exercise (LoRaWAN)

Team: YOUR NAME SURNAME + TEAMMATE NAME SURNAME

Person codes: PERSONCODE1 + PERSONCODE2

Date: 30 April 2026

EQ1 - Largest SF with packet success rate $\geq 75\%$

Assumptions: EU868, BW=125 kHz, payload=20 B, coding rate 4/5, explicit header, CRC on, single gateway, all 40 nodes use the same SF, and uncoordinated transmissions (pure-ALOHA overlap model). Total offered traffic $\lambda = 40 * 2 / 60 = 1.333$ packets/s. For packet airtime T, success probability is approximated as $P_{\text{success}} = \exp(-2 * \lambda * T)$.

SF	Airtime [ms]	P_success
7	56.576	0.860
8	102.912	0.760
9	185.344	0.610
10	370.688	0.372
11	741.376	0.138
12	1318.912	0.030

Answer EQ1: the largest spreading factor that still guarantees at least 75% success is **SF8** ($P_{\text{success}} \sim 0.760$). SF9 drops to ~ 0.610 , below the 75% target.

EQ2 - Best corrective action in field deployment

Selected action: 2) Move the nodes closer to the gateway.

Reason: the measured behavior is heterogeneous (some nodes good, some poor), which is typical of link-budget and radio-coverage differences across positions. Bringing weak nodes closer improves SNR/path loss and reduces packet loss for the worst nodes. Increasing SF would increase airtime and collision probability for everyone, while decreasing node count is not a deployment fix and does not directly address location-dependent weak links.